

```
!wget https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.csv
```

```
--2026-06-04 13:13:32-- https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.csv
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.19
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.1
HTTP request sent, awaiting response... 200 OK
Length: 9729 (9.5K) [text/plain]
Saving to: 'tips.csv'
```

```
tips.csv          100%[=====>]    9.50K  --.-KB/s    in 0s
2026-06-04 13:13:32 (85.4 MB/s) - 'tips.csv' saved [9729/9729]
```

```
from pyspark.sql import SparkSession

spark = SparkSession.builder.appName("BigDataAnalysis").getOrCreate()

df = spark.read.csv("tips.csv", header=True, inferSchema=True)

df.show(5)
df.printSchema()

print("Total Records:", df.count())

df.describe().show()
```

```
+-----+-----+-----+-----+-----+-----+
|total_bill| tip| sex|smoker|day| time|size|
+-----+-----+-----+-----+-----+-----+
|    16.99|1.01|Female| No|Sun|Dinner| 2|
|    10.34|1.66| Male| No|Sun|Dinner| 3|
|    21.01| 3.5| Male| No|Sun|Dinner| 3|
|    23.68|3.31| Male| No|Sun|Dinner| 2|
|    24.59|3.61|Female| No|Sun|Dinner| 4|
+-----+-----+-----+-----+-----+-----+
only showing top 5 rows
```

```
root
|-- total_bill: double (nullable = true)
|-- tip: double (nullable = true)
|-- sex: string (nullable = true)
|-- smoker: string (nullable = true)
|-- day: string (nullable = true)
|-- time: string (nullable = true)
|-- size: integer (nullable = true)
```

Total Records: 244

```
+-----+-----+-----+-----+-----+-----+
|summary| total_bill| tip| sex|smoker| day| time|
+-----+-----+-----+-----+-----+-----+
| count|      244|    244| 244| 244| 244| 244|
| mean|19.785942622950824|2.9982786885245902| NULL| NULL|NULL| NULL|
| stddev| 8.902411954856857|1.3836381890011815| NULL| NULL|NULL| NULL|
```

min	3.07	1.0	Female	No	Fri	Dinner
max	50.81	10.0	Male	Yes	Thur	Lunch

```
!wget https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic
```

```
--2026-06-04 13:17:32-- https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.153
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.153:443:
HTTP request sent, awaiting response... 200 OK
Length: 60302 (59K) [text/plain]
Saving to: 'titanic.csv'
```

```
titanic.csv          100%[=====>]  58.89K  --.-KB/s    in 0.001s
```

```
2026-06-04 13:17:33 (4.33 MB/s) - 'titanic.csv' saved [60302/60302]
```

```
df = spark.read.csv("titanic.csv", header=True, inferSchema=True)
```

```
df.show(5)
```

```
df.describe().show()
```

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch
1	0	3	Braund, Mr. Owen ...	male	22.0	1	0
2	1	1	Cumings, Mrs. Joh...	female	38.0	1	0
3	1	3	Heikkinen, Miss. ...	female	26.0	0	0
4	1	1	Futrelle, Mrs. Ja...	female	35.0	1	0
5	0	3	Allen, Mr. Willia...	male	35.0	0	0

```
only showing top 5 rows
```

summary	PassengerId	Survived	Pclass
count	891	891	891
mean	446.0	0.3838383838383838	2.308641975308642
stddev	257.3538420152301	0.48659245426485753	0.8360712409770491
min	1	0	1
max	891	1	3

```
# Total Passengers
```

```
print("Total Passengers:", df.count())
```

```
# Survivors Count
```

```
df.groupBy("Survived").count().show()
```

```
# Gender Count
```

```
df.groupBy("Sex").count().show()

# Passenger Class Count
df.groupBy("Pclass").count().show()

# Average Fare
from pyspark.sql.functions import avg

df.select(avg("Fare")).show()
```

Total Passengers: 891

```
+-----+-----+
|Survived|count|
+-----+-----+
|      1|   342|
|      0|   549|
+-----+-----+
```

```
+-----+-----+
|   Sex|count|
+-----+-----+
|female|   314|
|  male|   577|
+-----+-----+
```

```
+-----+-----+
|Pclass|count|
+-----+-----+
|      1|   216|
|      3|   491|
|      2|   184|
+-----+-----+
```

```
+-----+
|      avg(Fare)|
+-----+
|32.2042079685746|
+-----+
```

Check Missing Values

```
from pyspark.sql.functions import col, when, count

df.select([
    count(when(col(c).isNull(), c)).alias(c)
    for c in df.columns
]).show()
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|PassengerId|Survived|Pclass|Name|Sex|Age|SibSp|Parch|Ticket|Fare|Cabin|E
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|          0|        0|        0|  0|  0|177|        0|        0|        0|    0|  687|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
```

